

ABDUL KALAM MUSTAQ

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GEOSPATIAL DATA ANALYST AND URBANIST | ENVIRONMENTAL PLANNER

Geospatial Data Analyst and Urban Planner with 3+ years of experience in climate resilience, risk modeling, and GIS-based spatial and hazard assessment. Specialized in Spatial Analytics, Python programming, and Machine Learning for geospatial data processing. Expert at integrating **Nature Based Solutions (NbS)**, flood modeling, and data-driven decision-making for climate adaptation. Strong communicator with experience in Public Outreach, Climate Resilience Policy, and Strategic Evaluation. Currently pursuing a dual Master's in City Planning and Public Informatics, specializing in **Housing Equity, Systems Thinking, and Community-Centric Climate Adaptation Strategies**.

TECHNICAL PROFICIENCIES

GIS & Spatial Analysis: ArcGIS, QGIS, Google Earth, PostGIS, Coordinate System Transformations

Remote Sensing & Climate Modeling: NDVI Analysis, Land Use/Land Cover Classification, EPA SWMM, Hydrological Impact Modeling

Python for Geospatial Data Science: GeoPandas, Rasterio, GDAL, Pyproj, NumPy, Pandas, Xarray, HyRiver

Machine Learning for Risk Assessment: Scikit-Learn, K-Means Clustering, DBSCAN, TensorFlow, Predictive Modeling, Random Forest, Long Short-Term Memory (LSTM), Convolutional Neural Networks (CNN).

Big Data & Cloud GIS: Google Earth Engine, OpenStreetMap API, US Census API, PUMS Datasets

Data Visualization & Automation: Tableau, Power BI, ArcGIS StoryMaps, Jupyter Notebooks

Scientific Communication: Policy Briefs, Research Reports, Stakeholder Engagement, Public Speaking

Urban Planning & Policy: Urban NbS Policy Development, Climate Resilience, Stakeholder Engagement, Transportation Demand Management (TDM), Affordable Housing Development, Equity-Driven Planning, Climate Resilience & Adaptation Strategies, Environmental Justice, Regulatory Compliance, Policy Analysis & Implementation, 2D Drafting and 3D Modelling.

ACADEMIC CREDENTIALS

Rutgers University (Expected)

- **Master of Public Informatics** | GIS & Spatial Data Analysis
- **Master of City Planning** | Environmental & Climate Change Planning

Anna University (2016-2021)

- **Bachelor of Architecture** | Architecture & Urban Design

PROFESSIONAL EXPERIENCE

Rutgers Center on Law, Inequality, and Metropolitan Equity (CLiME), New Jersey

Geospatial Research Assistant

January 2025 - Present

- Conducts spatial research on **housing affordability and rent-burdened populations** in New Jersey using ArcGIS & Python.
- **Automated data pipelines** to streamline the extraction and transformation of datasets related to housing certificates of occupancy, building permits, and population demographics.
- Applied automation and geospatial techniques to enhance the **efficiency, accuracy, and scalability** of the analysis process.
- Generated detailed reports for publication with **data-driven insights**, highlighting key affordability challenges and policy implications.

Sponge Collaborative, India

Geospatial and Climate Change Analyst

January 2023 - July 2024

- Led hydrological GIS modeling for Chennai's flood resilience using ArcGIS, QGIS, and SWMM.
- Developed GIS-based processing tools for high-resolution urban flood risk mapping, integrating various datasets like land use landcover mapping, building footprints, drainage network, topography, groundwater recharge potential, etc.
- Conducted **spatial and environmental data analysis** for a **360-acre neighborhood**, enhancing **groundwater infiltration by 13%**.
- Designed flood mitigation strategies that improved resilience from **2-year to 5-year RP floods** across a **180-acre** flood-prone region.
- Co-authored the **Sponge Park Handbook**, guiding the prioritization of **1,000 sponge parks** using a spatial framework modeling.
- Conducted carbon sequestration analysis with TU Berlin, quantifying **103.3 tons of CO2 storage potential** using the **MRV system**.
- Built **multi-scalar geospatial models** for stormwater storage and groundwater recharge prediction.
- **Assessed institutional gaps** and conducted **Spatial, Environmental, and Social Analysis** using ArcGIS, AutoCAD, Excel, and field surveys in flood-prone areas of Chennai to implement and maintain **100 Sponge Parks** to mitigate flooding.

PROJECT EXPERIENCE

Flood Vulnerability Mapping in Woodbridge Township, New Jersey [Flood Vulnerability Mapping in Woodbridge Township.pdf](#)

- Developed GIS-based risk assessment models for identifying flood-prone residential zones using ArcGIS.
- Identified **163 land parcels** in danger of future flooding due to sea-level rise and inundation.
- Mapped **93.8% of critical wetlands** under extreme commercial and residential development pressure.
- Conducted **climate impact scenario analysis** to evaluate mitigation strategies for urban planning

Sponge Park Framework for Resilient Open Spaces [GCC Integrated Sponge Park \(DPR\).pdf](#)

- Co-authored a **Comprehensive Manual/Handbook** to guide decision-makers in implementing Sponge Parks with a **₹250 Crore** budget.
- Developed **Geospatial Models** in ArcGIS to identify **1,000 sponge parks** based on ecological, social, and financial criteria.
- Quantified the impact of BGI using **SWMM Hydrology Models** to find a **2-3 times** increase in soil storage capacity and **15% reduced runoff**.
- Conducted **Cost-Benefit Analysis** to identify **42% NPV** comparing green and gray infrastructure to mitigate five and 25-year RP storms.

CERTIFICATION

NEBDHub & NSDC Transportation Data Science Project (TDSP) [NSDC Analysing Crash Rates in NYC.pdf](#)

- Analyzed NYC OpenData transportation dataset to understand **crash patterns in NYC City**, covering data from **2012 to 2024**.
- Completed **six milestones using Python**, including data cleaning, time-series analysis, and developing critical thinking skills.
- Identified **11 high-risk intersections** for targeted safety improvements and provided recommendations to **NYCDOT**.

LIST OF PUBLICATIONS

- **"Impact of Proximity to Pedestrian Heavy Areas on Crashes in Midtown Manhattan, NYC"**
Link to the Report (completed in 2021):  [NSDC Analysing Crash Rates in NYC.pdf](#)
- **"An Urban Theater - Third Space in Urban Neighbourhoods"** - Undergraduate Thesis research and design report, Anna University.
Link to the Report (completed in 2021):  [An Urban Theater - Third Space in Urban Neighbourhoods.pdf](#)
- **"Reimagining OSRs for a Climate Resilient Coimbatore"** - A DPR for Coimbatore City Municipal Corporation.
Link to the Report (completed in 2022):  [Reimagining OSR's for a Climate Resilient Coimbatore \(DPR\).pdf](#)
- **"Kilambakkam Climate Park and Archaeological Interpretation Center"** - A DPR for Chennai Metropolitan Development Authority.
Link to the Report (completed in 2022):  [Kilambakkam Climate Park and Archaeological Interpretation Center \(DPR\).pdf](#)
- **"ISWD Sponge City Park Feasibility Report"** - A DFR for Greater Chennai Corporation.
Link to the DFR 1 (completed in 2023):  [GCC ISWD Sponge City Park Detailed Feasibility Report \(DFR\) 1.pdf](#)
Link to the DFR 2 (completed in 2023):  [GCC ISWD Sponge City Park Detailed Feasibility Report \(DFR\) 2.pdf](#)
- **"Integrated Sponge Park - Pilot"** - A DPR for Greater Chennai Corporation.
Link to the Report (completed in 2023):  [GCC Integrated Sponge Park \(DPR\).pdf](#)
- **"Integrated Community Welfare Center and Market"** - A DPR for Chennai Metropolitan Development Authority.
Link to the Report (completed in 2024):  [Integrated Community Welfare Center and Market \(DPR\).pdf](#)
- **"Sponge Park Framework for Resilient Open Spaces"** - A Sponge Park Handbook for Greater Chennai Corporation.
Link to the Report: *(yet to be published)*

COMPETITION HONORS AND NOMINATIONS

Winner, Chennai Lakefront Rejuvenation Competition, CMDA | 2023

Nominated, National Awards for Excellence in Architectural Thesis 2021, COA, 2022

Nominated, JK AYA Best Architecture Student of The Year Award 2021, JK Cement, 2022

Nominated, Thesis of the Year Awards 2021, ARCASIA, 2021

Shortlisted Top 25, Thesis of the Year Award 2021, Archmello, 2021

Finalist, Telangana State Urban Center of Excellence, NIUA, 2021

Asia Young Designer Awards (AYDA), Ethos, 2021

Finalist Top 15, Notions of India, TATA Steel, 2021

Shortlisted Top 10, Kekarav Experience Center Competition, TRU Realty, 2020

Berlin Bohemian Hostel, Archasm, 2019

Shortlisted Top 30, Plug and Play, SQR Factor, 2019

Finalist, ANDC Trophy 2017, ZONASA, 2017

PROFESSIONAL RECOGNITION

- **Council of Architecture, India**
Licensed Architect: CA/2023/158480